LESSON PLAN

BASIC RADIO & AVIONICS (25)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Purpose and operation of various safety devices used in electrical system of the aircraft.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Purpose and operation of various safety devices used in electrical system of the aircraft. Safety precautions while working on electrical systems in aircraft.
5. Specific Objectives	25.1.2.1 Trainee will be able to know and understand the constructional details of starters, ignition units and principles of. operation of various engine instruments

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of circuit breaker reset mechanism and fuse replacement procedure with instructor supervision
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a circuit breaker in an aircraft electrical system?
(b) Why are safety precautions critical when working on aircraft electrical systems?
(c) Can you list three common safety devices found in aircraft electrical systems?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Purpose of circuit breakers • Operation of thermal circuit breakers • Fuse types and ratings • Arc suppression devices • Ground fault protection	L & Ds	PPT, IB	15 min
(b)	• Safety precautions for electrical work • Lockout/tagout procedures • Personal protective equipment requirements • Handling high voltage components • Disposal of damaged electrical parts	L & Ds	PPT, PDF	20 min
(c)	• Review of safety device failures • Impact of electrical faults on aircraft systems • Maintenance intervals for safety devices • Troubleshooting common electrical issues • Importance of regular inspections	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a thermal circuit breaker and a magnetic circuit breaker.
(b) Describe the correct procedure for replacing a blown fuse in an aircraft electrical system.
(c) List three safety precautions that must be followed before working on aircraft electrical systems.

12. Home Assignment and Follow Up	Prepare a brief report on the importance of regular inspection of safety devices in aircraft electrical systems, referencing the next lesson plan (LP 02) on starters and ignition units.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.
6. Are there specific aircraft models where the safety devices differ significantly?

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min